

$$1. -\left(\frac{5}{12} \log_2 \frac{5}{12} + \frac{7}{12} \log_2 \frac{7}{12}\right) \approx 0.980$$

2. x_1 : divide ^{by} into 1.5, 2.5, 3.5, 4.5

threshold 1.5 $x_1 < 1.5$ ($\frac{4}{12}$, 2Y, 2N) $H = \frac{4}{12} \times 1.0 + \frac{8}{12} \times 0.954$

$x_1 > 1.5$ ($\frac{8}{12}$, 3Y, 5N) $= 0.969$

2.5 $x_1 < 2.5$ ($\frac{6}{12}$, 2Y, 4N)

$x_1 > 2.5$ ($\frac{6}{12}$, 3Y, 3N)

3.5 $x_1 < 3.5$ ($\frac{7}{12}$, 2Y, 5N)

$x_1 > 3.5$ ($\frac{5}{12}$, 3Y, 2N)

4.5 $x_1 < 4.5$ ($\frac{10}{12}$, 4Y, 6N)

$x_1 > 4.5$ ($\frac{2}{12}$, 1Y, 1N)

$x_T = 1.5$ $H_1 = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = 1$

$H_2 = -\left(\frac{3}{8} \log_2 \frac{3}{8} + \frac{5}{8} \log_2 \frac{5}{8}\right) = 0.954$

$H_w = \frac{4}{12} \times 1 + \frac{8}{12} \times 0.954 = 0.969$

$IG = 0.011$

$x_T = 2.5$ $IG = 0.021$

$x_T = 3.5$ $IG = 0.072$

$x_T = 4.5$ $IG = 0.004$

So use $x_T = 3.5$ as the final threshold

$x_2 : N = \frac{5}{12}$ All N in y

$Y = \frac{7}{12}$ 2 N 5 Y

$H_2 = \frac{5}{12} \times 0 + \frac{7}{12} \left(-\frac{2}{7} \log_2 \frac{2}{7} - \frac{5}{7} \log_2 \frac{5}{7}\right) = 0.503$

$IG_2 = 0.477$

$x_3 : H_3 = \frac{6}{12} \cdot 0.918 + \frac{6}{12} \times 1.00 = 0.959$

x_2 is the first

$IG = 0.021$

attribute, $IG = 0.477$